Geophys. J. Int. (2000) **143,** 22–38

Palaeomagnetism, rock magnetism and geochemistry of Jurassic dykes and correlative redbeds, Massachusetts, USA

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Accepted 2000 March 31. Received 2000 March 20; in original form 1998 October 20

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SUMMARY

Jurassic diabase dykes, sills and sedimentary rocks in central Massachusetts were sampled for palaeomagnetic analysis. The intrusions fall into three of the chemical types for eastern North American diabases: high TiO_2 quartz-normative (Holden); low TiO_2 quartz-normative (Ware); and high Fe_2O_3 quartz-normative (Pelham–Loudville). The characteristic magnetizations in the majority of intrusive samples unblock between 550 °C and 580 °C, with Curie temperatures in a discrete interval between 556 °C and 580 °C. The dominant remanence in the diabases is carried by C1 to C3 oxidation-exsolved titanomagnetite occurring as euhedral grains, as fine needles or dust in the matrix, as devitrifed glass, and as fine magnetite-ilmenite-silicate symplectite. In some dykes, titanomagnetite was further modified by deuteric oxidation during post-magmatic cooling, creating titanomaghematite and/or a granulation of the magnetite. Palaeopoles for the three diabase groups are: Holden, 60.1°N, 80.5°E, $A_{95}=4.1°$; Ware, 73.5°N, 85.8°E, $A_{95}=3.9°$; and Pelham–Loudville, 65.3°N, 95.6°E, $A_{95}=4.1°$.

These data are combined with samples from two stratigraphic sections through the Early Jurassic part of the Sugarloaf Formation in the Deerfield Basin representing both fine-grained mudstones and coarser arkoses. These haematite-dominated rocks reveal several components of magnetization, a steep recent field direction, an intermediate secondary diagenetic overprint direction in both mudstones and arkoses, and a high-temperature shallow primary direction found only in the mudstones. Palaeopoles for the Sugarloaf Formation are: mudstones, 57.7°N, 81.3°E, $A_{95}=9.1°$; and arkoses, 75.1°N, 131.6°E, $A_{95}=5.9°$. Based on the new palaeomagnetic data reported here, the North American plate in the Middle Jurassic was at higher palaeolatitudes than indicated by the present North American apparent polar wander path.

Key words: APW (apparent polar wander), Jurassic, mineralogy, North America, palaeomagnetism

1 INTRODUCTION

The accurate definition of an apparent polar wander path (APWP) for North America is crucial to our understanding of plate tectonics, particularly during the Jurassic when major rifting and continental reorganization occurred. Unfortunately, the Jurassic part of the APWP for North America is poorly understood for several reasons (Hagstrum 1993; Courtillot *et al.* 1994). There are abundant data for North America during the Late Triassic–Early Jurassic, but they are derived predominantly from two locations, namely the Colorado Plateau and Mesozoic basins of the Newark Supergroup found along the East Coast. Results on units of similar age from the two locations are in poor agreement, providing evidence both for and against the J1 cusp, a sharp hairpin turn in the APWP in the Early Jurassic. Both regions yield pole positions resulting from complex magnetizations, for which problems of remagnetization, overprints, tectonic movements, and poorly defined age assignments are common. In addition, some studies indicate the presence of high-latitude poles (Van Fossen & Kent 1990; Kodama & Mowery 1994) in the mid-Jurassic, requiring alteration in proposed APWPs (May & Butler 1986; Bazard & Butler 1994). Considering palaeopoles from other Atlantic-bordering continents, both Van der Voo (1992) and Courtillot *et al.* (1994) have suggested that the North American path during the Jurassic should be at a higher latitude than originally proposed.

Considerable palaeomagnetic work has been undertaken recently in eastern North America (McIntosh *et al.* 1985; Witte & Kent 1990, 1991; Witte *et al.* 1991; Kent *et al.* 1995; Kent & Olsen 1997; Kodama & Mowery 1994; Kodama *et al.* 1994a, 1994b), with attention given to both sedimentary and interbedded volcanic units of the Newark Supergroup (Olsen 1978,

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1997). These continental sediments ('redbeds') and extrusive volcanic rocks accumulated from the Late Triassic to Early Jurassic in a series of extensional basins as a result of initial separation between North America, Africa and Europe. All Early Jurassic units sampled have normal polarities, while the older (Late Triassic) sedimentary rocks record both normal and reversed polarities.

Biostratigraphic ages of the Newark Supergroup sediments range from Carnian (Mid-Triassic) to Pliensbachian (early mid-Jurassic) (Cornet 1977; Olsen *et al*. 1982). The major extrusive activity occurred during the Hettangian, and, using Milankovitch cycles observed in lake sediments, Olsen (1986) has suggested that the lava flows were emplaced rapidly, representing one volcanic event. Precise age determinations for igneous units in the Newark Supergroup has long been a problem (Sutter 1988; Seidemann *et al*. 1984), with a wide range of measured radiometric ages available in the literature. There are generally believed to be two major volcanic episodes represented, one at 195 Ma and another at 175 Ma (Sutter & Smith 1979), but recent work has indicated that the earlier date may be too young and that the later date may be related to hydrothermal activity (Sutter 1988; Dunning & Hodych 1990). The idea that all the Newark extrusive rocks and dykes represent only a short emplacement time is counteracted by observations of numerous dyke swarms with cross-cutting relations (Dooley & Smith 1982), and the recognition of chemically distinct igneous provinces in many of these areas (De Boer *et al*. 1988; McHone 1992).

Although a number of pioneering studies (Du Bois *et al*. 1957; Irving & Banks 1961; De Boer 1968) were done on rocks of the Hartford–Deerfield basins in southern New England, all were hampered by limited to non-existent demagnetization. Even the detailed work by Smith & Noltimier (1979) on intrusive igneous rocks in Connecticut suffers from insufficient demagnetization techniques. Work by Prevot & McWilliams (1989) on extrusive rocks from the Deerfield, Hartford, and Newark basins offers new clues about the magnetic characteristics of these rocks, but too few sites were evaluated to enable an accurate definition of palaeomagnetic vectors. In this study we present palaeomagnetic and rock-magnetic data from diabase intrusions sampled outside the Mesozoic basin proper in central Massachusetts, as well as limited palaeomagnetic data from lacustrine mudstones and redbeds of earliest Jurassic age from the Deerfield Basin. The analyses include petrography and geochemistry of the intrusions, as well as detailed rock-magnetic studies on both igneous and sedimentary samples, in order fully to characterize the recorded magnetizations.

2 GEOLOGICAL SETTING

Numerous intrusions of Mesozoic age are found in eastern North America, and those with quartz tholeiitic composition are assigned to the Eastern North America (ENA) igneous province (Weigand & Ragland 1970). In central Massachusetts, numerous quartz tholeiitic dykes and sills (Fig. 1) intrude deformed sedimentary and igneous rocks of Late Precambrian to Devonian age (Robinson *et al*. 1992; Zen *et al*. 1983). These dykes extend in a NE–SW-trending en echelon pattern for 175 km from the Connecticut border to the New Hampshire border, and are found on either side of, but not within, the Mesozoic Hartford and Deerfield basins. The dykes were initially grouped into four sets based on outcrop location and orientation: the Loudville set, west of the rift basins; the Pelham set, close to, but east of, the rift basins; the Ware set at an intermediate distance to the east; and the Holden set, the farthest east (Fig. 1). Refinements on the dyke grouping, including the merging of the Pelham and Loudville sets, was done later on the basis of distinct geochemical signatures. No published radiometric data exists for any of the Jurassic dykes in Massachusetts.

In the Deerfield basin three sedimentary units and one igneous extrusive unit are recognized: Sugarloaf Arkose, Deerfield Basalt, Turners Falls Sandstone and Mount Toby Conglomerate (Robinson & Luttrell 1985). The Sugarloaf Arkose, the oldest of the units, consists of 2000 m of fluvial and alluvial-fan deposits, with the uppermost part of the unit comprising 9 m of lacustrine deposits (Stevens & Hubert 1980). Based on pollen and spore identification by Cornet (1977), the Triassic–Jurassic boundary lies just below this lacustrine layer. Recently, these rocks have been informally named the Fall River beds (Olsen *et al*. 1989) and incorporated into regional syntheses (Weems & Olsen 1997). Facies found in the lacustrine section are thin-bedded grey to red mudstones, laminated grey mudstones and greyish red to grey sandstones and fine arkoses, described in detail by Stevens & Hubert (1980). The interstratification of these facies records the advances and retreats of shallow perennial lakes on the valley floor. The red colouring of the rocks is derived from haematite coatings on the detrital grains produced by dehydration of limonite staining present at the time of deposition (Hubert & Reed 1978; Stevens & Hubert 1980). Many of the sandstones have been bleached by later hydrothermal activity (Hubert & Dutcher 1999) with the addition of some late iron oxide deposition from the dissolution of Fe-silicates. Strata in the Deerfield Basin currently dip at low to moderate angles ($10^\circ - 20^\circ$) to the east, due to basin subsidence during deposition of the continental sediments. The eastern boundary of the basin, the Connecticut Valley border fault, is a west-dipping listric normal fault with approximately 5 km of vertical displacement in northern Massachusetts increasing to 8 km in southern Massachusetts (Robinson *et al*. 1986). There is no field evidence for horizontal rotation of blocks within the basin. Several normal faults of Mesozoic age are recognized east of the Deerfield Basin in the metamorphic highlands, but field evidence indicates that only small displacements have occurred (Ashenden 1973; Robinson *et al*. 1986).

Samples for both chemical–petrological studies and palaeomagnetic measurements were collected together in the field using either a portable gasoline-powered drill or as hand samples, and orientated with a Brunton compass, and whenever possible a sun compass. Dyke exposures vary in thickness from less than 1 m up to 55 m, allowing samples to be collected from both chilled margins and well-crystallized interiors. Except for the Loudville dykes, exposed west of the Hartford Basin, all palaeomagnetic diabase sites are located east of the Connecticut Valley border fault in the metamorphic highlands, but are distributed spatially over the known outcrops (Fig. 1).

Sedimentary rock samples were collected in three stratigraphic sections at the north end of the Deerfield Basin just east and northeast of Greenfield, Massachusetts (stars in Fig. 1). All sections are immediately below the Deerfield Basalt. Both red mudstones and paler fine-grained sandstones and arkoses (herein referred to simply as arkoses) were sampled in each section

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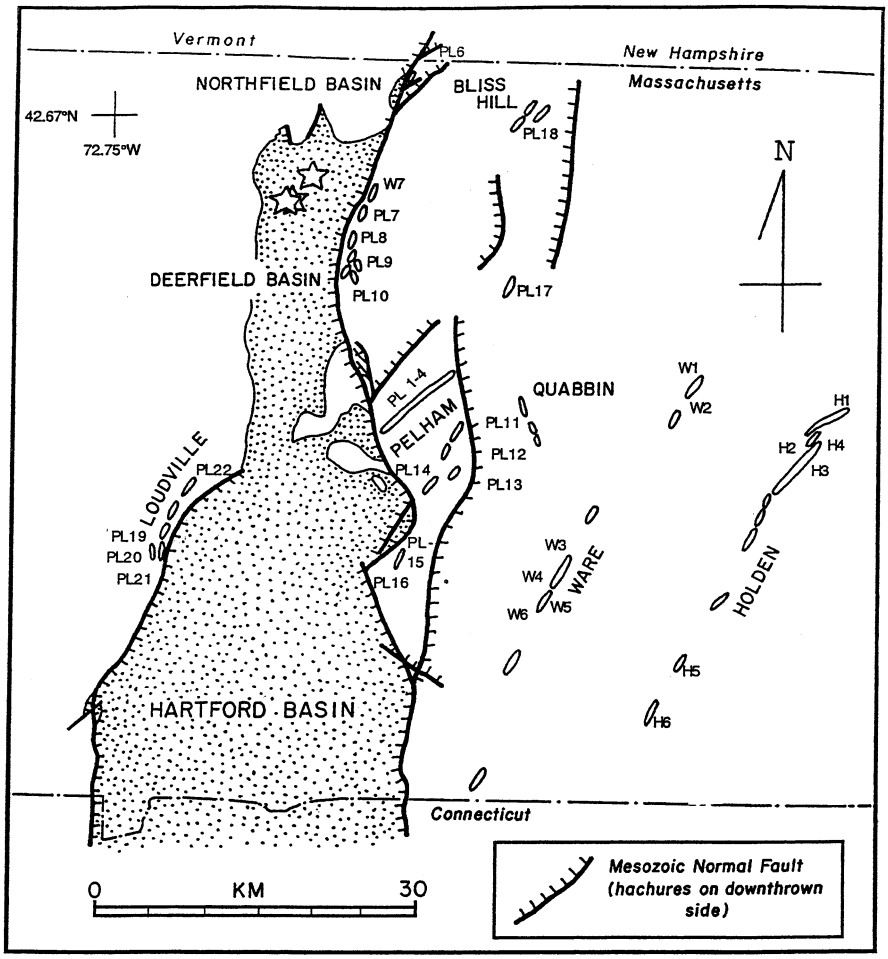


Figure 1. Map of central Massachusetts showing the locations of the Mesozoic basins, and dykes and sills intruded into crystalline rock. Locations of the palaeomagnetic sites from the Holden, Pelham–Loudville and Ware diabase groups are labelled. Stars indicate locations of stratigraphic sections through the Fall River beds in the Deerfield Basin.

3 PETROGRAPHY AND GEOCHEMISTRY

Sedimentary and diabase samples were examined by reflected-light microscopy to identify the opaque phases, morphologies, oxidation-exsolution textures, grain size, and degree of alteration for the purpose of evaluating the nature of the magnetic carriers and timing of remanence. A brief overview of diabase petrography and geochemistry is given here to show the distinct nature of the diabase groups and to compare the central Massachusetts diabases with the regional eastern North American (ENA) diabase suite.

3.1 Petrography of sedimentary rocks

Optical examination of selected samples of mudstones and arkoses reveals several forms of iron oxides, as has been reported by other workers in the Connecticut Valley (Hubert & Reed 1978; Stevens & Hubert 1980; Hubert *et al.* 1992). Most obvious in the finer-grained rocks is ubiquitous haematite staining on the detrital grains. This is present as red (bright) rims around the grains and as amorphous material between grains. Opaque detrital grains are seen to be primarily haematite after magnetite (martite), identifiable by relic {111} ilmenite lamellae. Some detrital haematite is present, as well as a few rare unaltered detrital grains of magnetite (Fig. 2a). Scarce ilmenite has been further oxidized to haematite + rutile. Also observed are iron-silicate grains that have altered *in situ* to produce small well-formed crystal grains of haematite (Fig. 2b). The alteration of the silicates to haematite is more common in the mudstones than in the sandstones, which are predominately quartz and feldspar and have few sheet silicates present. The pale-red sandstones (arkoses) have considerably less haematite staining and interstitial iron oxide material than the mudstones.

3.2 Diabase petrography

The Holden diabases contain plagioclase, Mg-rich orthopyroxene, augite, and pigeonite phenocrysts. Pigeonite is locally mantled by augite, and glomerocrysts of clinopyroxene are common. The matrix contains plagioclase, clinopyroxene, biotite, and skeletal to cruciform to euhedral morphologies of titanomagnetite (Fig. 3). Matrix titanomagnetites have textural stages of C2 to C3 (Haggerty 1991). Samples from well-crystallized interiors of the dykes have {111} or sandwich oxidation-exsolution of ilmenite lamellae in the titanomagnetite. Some interior samples show a low-temperature maghematization or granulation superimposed on the oxidation-exsolution features (Fig. 3a). Ilmenite is present as discrete grains or as composite grains with magnetite, and most commonly oxidation-exsolution

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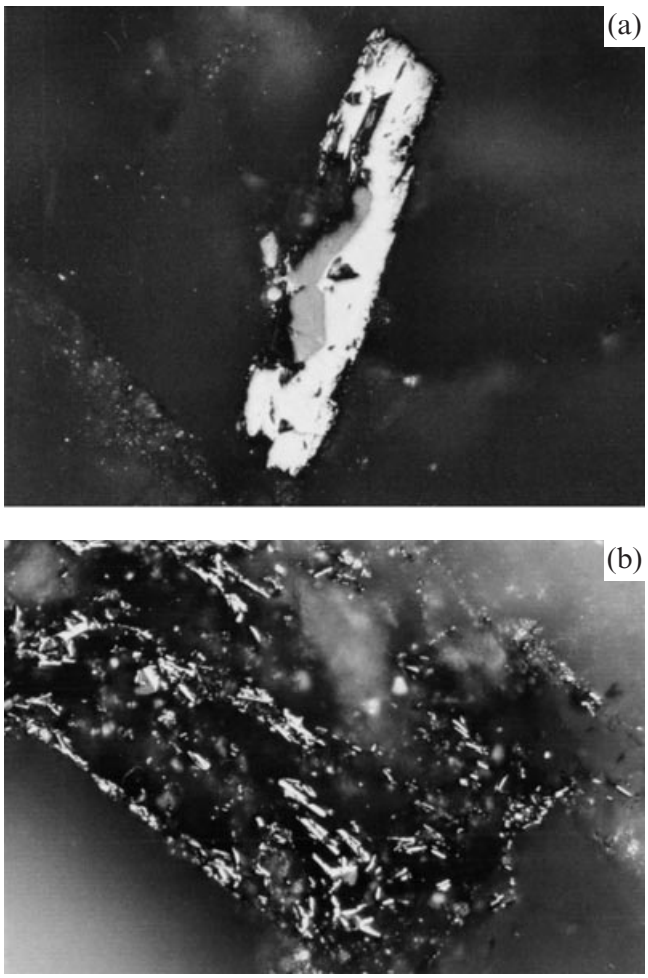


Figure 2. Reflected-light photomicrographs of mudstone (sample 324), in oil immersion and with a blue filter. (a) Detrital titanomagnetite grain with {111} ilmenite lamellae. Also note fine-grained haematite in the mudstone matrix. (b) Well-developed haematite needles forming from the breakdown of amphibole. The field of view for both photos is 74μ.

lamellae in magnetite. There are pyrite and titanomagnetite intergrowths, discrete pyrites, chalcopyrite and pentlandite, and late-stage pyrites surrounding titanomagnetites. Primary magnetite-ilmenite-feldspar symplectites and magnetite-ilmenite-spinel symplectites are found in numerous samples (Fig. 3b). Micrographic intergrowths of alkali feldspar and quartz are common in this group and rare in the samples from the other two diabase groups.

The Pelham–Loudville diabases are composed of plagioclase, augite and less commonly pigeonite, rare olivine usually as microphenocrysts rimmed by clinopyroxene, and discrete grains of ilmenite and titanomagnetite. Titanomagnetite (C1–C3) morphologies range from euhedral grains to skeletal arrangements with herringbone texture to fine beads of magnetite surrounding silicates, and, on an even finer scale, titanomagnetite dust. High-temperature oxidation features are present in dykes, showing oxidation textures formed above the magnetite-ulvöspinel solvus. Interestingly, in a subset of the Pelham–Loudville dykes, samples have a titanomagnetite that has not been oxidized until lower temperatures, as shown by a lit-par-lit texture in numerous skeletal and euhedral titanomagnetites (Fig. 3b). To form the lit-par-lit texture (Ramdohr 1969; Haggerty 1991), exsolution of ulvöspinel from magnetite is required and hence must take place below the consolute point of approximately 600°C (Buddington & Lindsley 1964). The exsolution of ulvöspinel was followed by a later oxidation altering the ulvöspinel to ilmenite. Based on textural evidence, this oxidation probably occurred during the initial cooling of the diabase following emplacement. There are minor amounts of a late amorphous Fe-Ti oxide, pyrite, chalcopyrite, and biotite, and trace amounts of secondary calcite, Fe- and Mg-chlorite, apatite and zircon.

Chilled margins of Ware diabases contain phenocrysts of plagioclase, and microphenocrysts of olivine rimmed by pigeonite or augite. Clinopyroxene phenocrysts are predominantly pigeonite, but augite is also present. The matrix is composed of plagioclase, pigeonite and augite, devitrified glass, straw-coloured oxidized glass and fine-grained herringbone skeletal titanomagnetite. Thin sections show less than 5 per cent alteration. Locally, titanomagnetite shows some granulation. Minor late-stage amorphous Fe-Ti oxide and rare discrete grains of ilmenite are present. Pyrite or chalcopyrite and biotite make up to 1 per cent of the rock, and associated with this are trace amounts of apatite, quartz, K-feldspar and calcite.

3.3 Diabase geochemistry

Samples were analysed by X-ray fluorescence (XRF) for major and trace elements at the Department of Geosciences, University of Massachusetts, thus permitting detailed comparisons of the central Massachusetts diabases with the eastern North American (ENA) diabase suite. Aliquants of powder sampled for XRF analyses were prepared following a modification of the methods of Norrish & Chappell (1967) and Norrish & Hutton (1969). 48 samples were analysed in duplicate, for major elements using fused glass disks, and for trace elements as pressed powders.

According to the major-element chemistry, these diabases are quartz-normative tholeiites, as are many of the ENA diabases. The magnesium values, on average, are similar to the values determined for the other ENA diabases but are significantly lower than in the Triassic dykes from Rhode Island (McEnroe 1995). Major elements clearly separate the 48 dyke analyses into three distinct groups. On a regional scale, the Holden, Pelham–Loudville and Ware diabase groups fall into three of the four chemical types for the ENA diabases (Fig. 4a) defined by Weigand & Ragland (1970). These are the high-TiO_2 quartz-normative (HTQ), the low-TiO_2 quartz-normative (LTQ), and the high-Fe_2O_3 quartz-normative (HFQ) types as shown in Fig. 4(a), in which the high-field-strength incompatible element Ti (as TiO_2) is plotted against the mafic index $Fe_2O_3/(Fe_2O_3+MgO)$. The Holden dykes plot in the HTQ field, but show a systematic variation in both TiO_2 and Fe_2O_3, a feature found in a number of other HTQ samples (Ragland *et al.* 1992). The Pelham–Loudville diabases plot in a tight cluster in the HFQ field. The high Fe content, averaging 16 wt. per cent Fe_2O_3, and high Ti content are characteristic of the Pelham–Loudville. These dykes are further distinguished by their low MgO content, namely $\sim$6 wt. per cent. In sharp contrast, the Ware diabases plot in the LTQ field and have a low Fe content, $<$12 wt. per cent Fe_2O_3, and a markedly lower Ti content, $\sim$0.80 wt. per cent. In addition to the low Fe and Ti contents of these dykes, they have a significantly higher MgO

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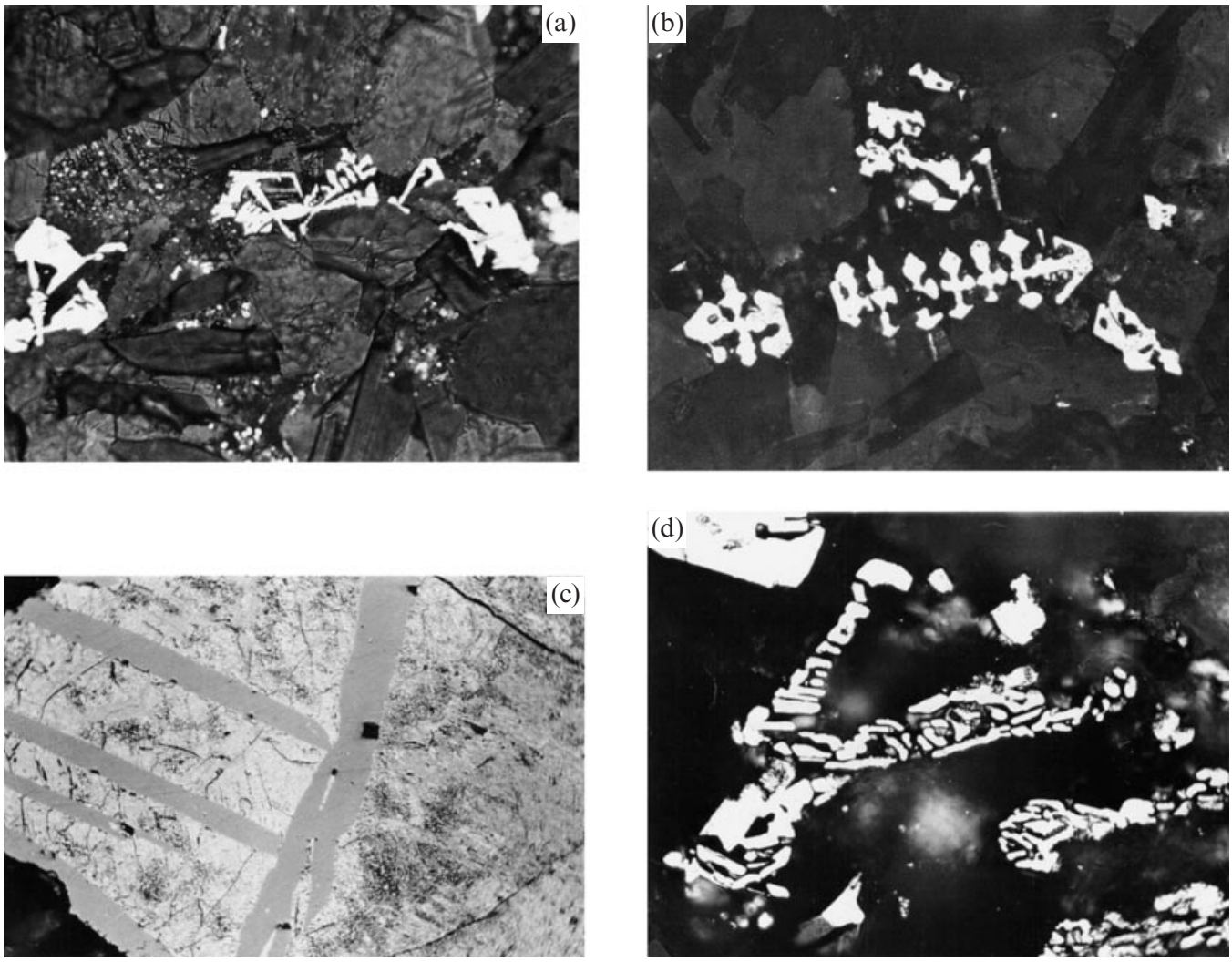


Figure 3. Reflected-light photomicrographs all photographed in oil immersion with a blue filter. (a) C1–C2 titanomagnetites with herringbone and cruciform to euhedral morphologies. Note the very fine titanomagnetite in the matrix. The field of view is 130μ. (b) Skeletal titanomagnetite grains with minor oxidation-exsolution features. Field of view is 63μ. (c) Large titanomagnetite grain with {111} sandwich lamellae of ilmenite. Superimposed on this is a later granulation in the magnetite-rich areas, shown as dark grey to black areas, and a low-temperature oxidation creating areas of maghemite as shown by curvilinear cracks. Field of view is 130μ. (d) Magnetite-ilmenite-silicate symplectites in matrix. Field of view is 130μ.

content, usually >7 wt. per cent. Although the Holden dykes show considerable variation in MgO, it is with a systematic linear trend varying from 6.5 to 8.5 wt. per cent MgO, probably controlled by crystal fractionation.

The distinctive nature of the three diabase groups is also borne out by the trace-element data (Figs 4b to d). Although the Holden dykes show a large range in major and trace elements, there is little overlap of these elements with the Pelham–Loudville or Ware diabases. The Holden dykes are richer in incompatible elements Nb, Sr, Ce, Zr, and compatible elements Cr and Ni than the Pelham–Loudville or Ware dykes. The three chemical groupings are well shown in a plot of Sr versus Y (Fig. 4b), with strong enrichment of Sr in the Holden dykes, whereas the Pelham–Loudville dykes are highly enriched in Y and poor in Sr. Although containing only moderate amounts of Y and Sr, the Ware dyke analyses are remarkably clustered and plot distinctly outside the two regions defined by the Holden and Pelham–Loudville analyses. The chemistry of Pelham–Loudville intrusions is extremely uniform throughout and is characterized by very high Fe, TiO_2, Y and V contents. As shown in Fig. 4(c), the high V and relatively enriched Zr contents separate the Pelham–Loudville dykes from the two distinct fields for the Ware and Holden dykes. A similar pattern is shown in Fig. 4(d), with Zr (p.p.m.) plotted against Y (p.p.m.) showing the three groupings, although in this diagram a linear fractionation trend is more evident in the behaviour of Zr and Y in the Holden diabases, as is also typical for major elements. In summary, based on major elements the Holden, Pelham–Loudville and Ware diabases fall broadly into the three fields of the quartz-normative ENA diabases as previously defined, and the distinctions are further refined by trace-element data.

3.4 Discriminant diagrams

The high-field-strength incompatible elements, Ti, Zr, Y and Nb, analysed in this study are commonly used discriminators for the assignment of tectonic settings of basaltic rocks (Pearce

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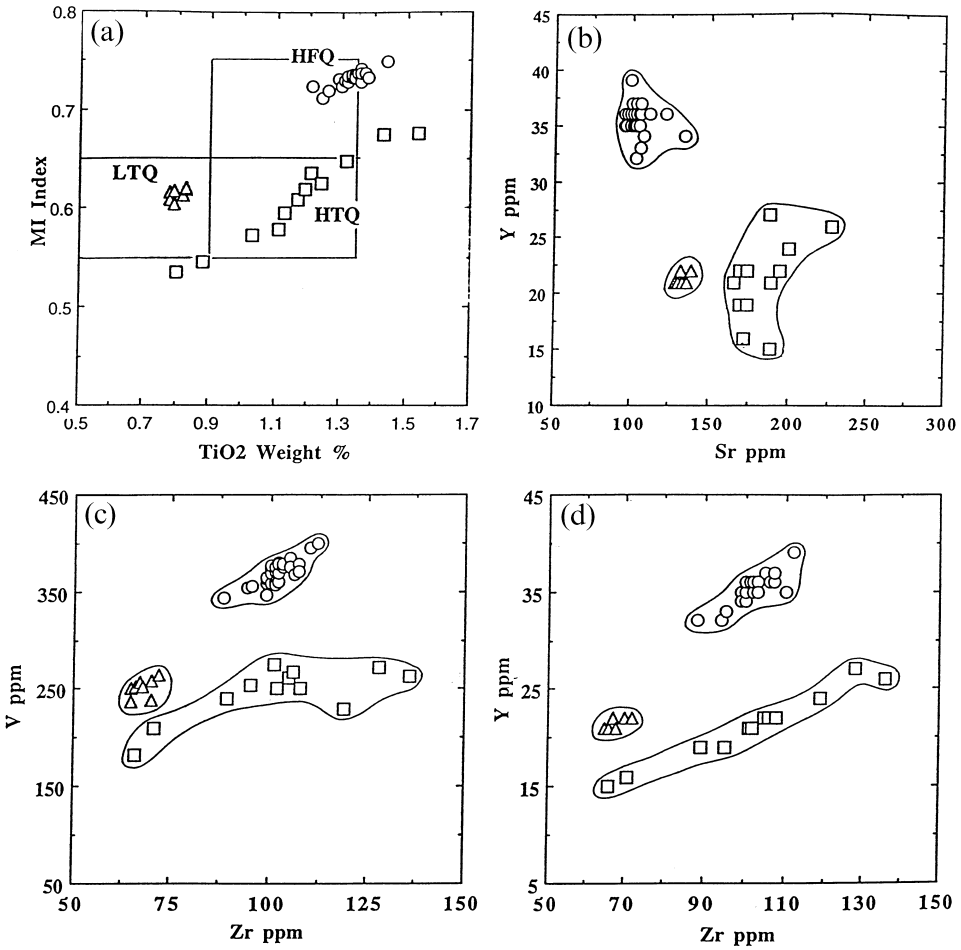


Figure 4. (a) Diagram of weight ratio of $Fe_2O_3/(Fe_2O_3+MgO)$ versus wt. per cent TiO_2. The three fields of low-TiO_2 (LTQ), high-TiO_2 (HTQ) and high-Fe_2O_3 quartz-normative (HFQ) diabases as defined by Weigand & Ragland (1970) are shown with diabase data. The three chemical groups are well displayed in trace-element plots (p.p.m.). (b) Sr plotted against Y. (c) V plotted against Zr. (d) Zr plotted against Y. Symbols are: Holden: open squares; Pelham–Loudville: open circles; Ware: open triangles.

& Cann 1971, 1973; Pearce & Norry 1979; Meschede 1986), and are used to discriminate the tectonic fields of mid-ocean-ridge basalts (MORB), island-arc basalts, low-K tholeiites, calc-alkaline basalts, and within-plate basalts. Although there is little question about the tectonic environment of these rocks, a rift-related continental setting, these diagrams provide important information about the changing environment of Pangea as it was breaking up. The tectonic relationships found on the Meschede diagram (Fig. 5), which uses the immobile trace elements Nb, Zr, and Y to discriminate among N-type MORBs, P-type MORBs, within-plate tholeiites, and within-plate alkali basalts, are also consistently found on other discriminant diagrams. The Holden dykes all plot in the field of within-plate tholeiites, whereas the Pelham–Loudville and the Ware diabases fall in the MORB field, although in this diagram to the more restricted N-type MORB. The discriminant diagrams show a clear distinction for each of the three diabase groups. Although we do not interpret the Pelham–Loudville and Ware diabases to be of strictly oceanic character, we do interpret this signature as probably reflecting a thinning of the continental crust, possibly as Pangea started to break up, as compared with the thicker continental crust present when the Holden diabases intruded at an earlier time as signified by their within-plate signature.

The distinct chemical natures of the Holden, Pelham–Loudville, and Ware groups are in good agreement with the petrographical differences found. Based on the petrography and geochemistry and their within-plate geochemical signature, we predict that the Holden diabases should have a pre-J1-cusp palaeomagnetic pole, reflecting the setting of a thicker continental crust into which these dykes must have intruded. If the Ware and Pelham–Loudville dykes plot just after the J1 cusp, then these intrusions could well represent the initial break-up of Pangea.

4 PALAEOMAGNETISM

4.1 Laboratory procedures

The natural remanent magnetization (NRM) of 254 diabase samples was measured on a Molspin spinner magnetometer at the University of Massachusetts (UMASS), Amherst, or on a Jr-5 9 Hz spinner magnetometer at the Norwegian Geological Survey (NGU) in Trondheim, Norway. For alternating field (AF) demagnetization treatment, specimens were progressively demagnetized up to a peak induction of 100 mT using a three-axis tumbling (at NGU) or a single-axis non-tumbling

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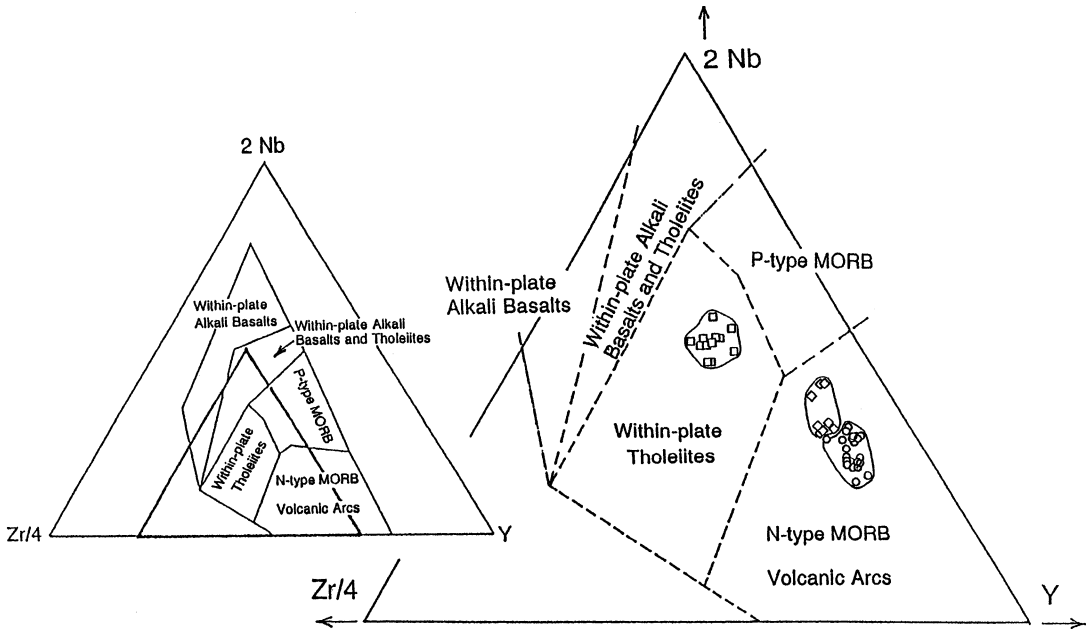


Figure 5. Discriminant diagram for the trace elements Nb, Zr, and Y (after Meschede 1986), indicating tectonic settings for tholeiities and all types of within-plate basalts. All diabases plot in the area of the enlarged insert. Symbols are: Holden: open squares; Pelham-Loudville: open circles; Ware: open diamonds.

Schoenstedt GSD-1 AF geophysical specimen demagnetizer (at UMASS). For thermal demagnetization, specimens were heated in air to a maximum temperature of 580 °C for the diabase samples using a non-inductively wound furnace, and cooled in a field-free chamber at NGU or UMASS. Sedimentary rock samples were measured on a cryogenic magnetometer at the University of Wyoming, with thermal demagnetization done in a shielded furnace in steps up to 660 °C on all samples.

Hysteresis properties (micromagnetometer or vibrating sample magnetometer, Princeton Applied Research), Curie temperatures (vibrating sample magnetometer with furnace), and temperature versus susceptibility measurements (Kappa bridge) were made at the Institute for Rock Magnetism, University of Minnesota. Hysteresis curves were measured at room temperature in a maximum field of 2 T. Magnetic susceptibility values were measured on a Sapphire susceptibility bridge at UMASS.

4.2 Remanence behaviour of the diabases

The natural remanent magnetization (NRM) of the Jurassic diabase samples shows some variation in directions and intensities, with initial magnetizations ranging from 0.3 to 4.0 A m^{-1}. Under AF demagnetization, mean destructive fields (MDFs) are between 20 and 40 mT (Fig. 6) for all three dyke groups. Most samples show two components of magnetization, with the secondary, low-coercivity direction being removed between 10 and 40 mT. The Holden dykes generally yield a small secondary component removed at low fields, and a higher-coercivity characteristic component above 20 mT. This component consistently has declinations east of north, and shallow to intermediate positive inclinations (Fig. 7c). The Pelham–Loudville group, containing the largest number of samples, has numerous minor secondary components, but most are removed at low fields (10–20 mT). The northerly declinations and steep inclinations of most of these secondary components

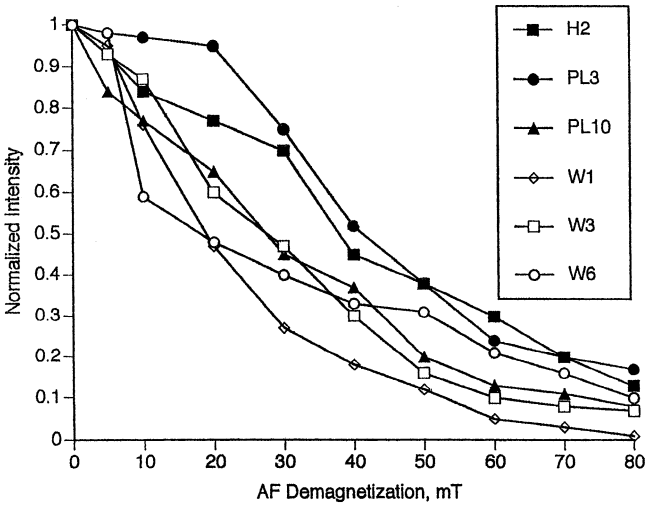


Figure 6. Normalized NRM demagnetization curves for representative AF samples from Holden (H), Pelham–Loudville (PL) and Ware (W) intrusions. MDFs vary from 20 to 40 mT for most samples.

point to a present-field overprint. At demagnetization levels above 20 mT, a stable characteristic component with a north-northeast declination and intermediate down inclination is found (Fig. 7d). The NRM values in this group are often higher than the values for the Holden or Ware diabases, reflecting the greater iron and magnetite content in Pelham–Loudville dykes. Ware diabase samples show a large west to north-westward secondary component, removed between 10 and 15 mT, above which a well-defined characteristic component is isolated (Fig. 7e). This direction is similar to that of the other dykes in its northerly declination, but consistently has steeper inclinations than the other two groups. The Ware samples show a large range in coercivity, with mean destructive fields

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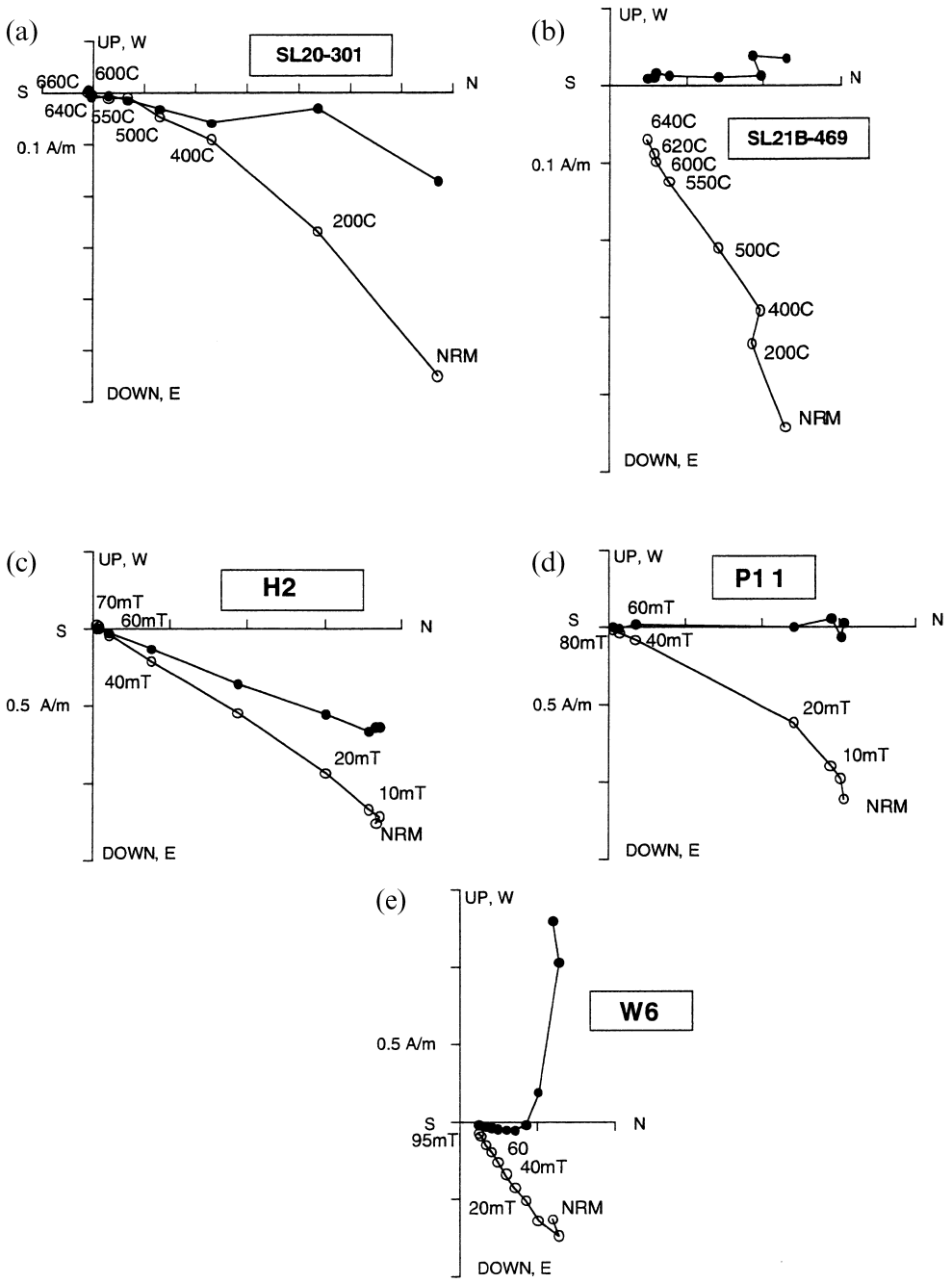


Figure 7. Vector end-point diagrams (Zijderveld 1967) for typical samples from the mudstone (SL20-301) and arkose (SL21B-469) of the Fall River beds, and the Holden (H2), Pelham–Loudville (PL11), and Ware (W6) diabases. All diagrams are plotted in *in situ* coordinates. Open/closed symbols are projections in the vertical/horizontal plane.

(MDFs) from 15 to 80 mT, but the high coercivities are limited to samples from chilled margins. Iron oxides in these samples are dominated by fine skeletal crystals and needles of titanomagnetite in the matrix, providing pseudo-single-domain (PSD) grains with a stable, high-coercivity remanence. Samples with lower MDFs (20–40 mT) have more multidomain-size (MD) magnetite combined with PSD-size tianomagnetite. Despite this wide range of magnetic decay, all samples show the same characteristic direction. Thermal demagnetization studies reveal a distribution of unblocking from 500 to 580 °C (Fig. 8) for diabases with titanomagnetite and magnetite carriers. A few samples from the interiors of the Holden and Ware dykes show a lower unblocking temperature, consistent with maghemite or a granulation to the magnetite. Characteristic remanent directions obtained with thermal demagnetization are similar to those obtained in alternating field studies.

4.3 Remanence behaviour of the sedimentary rocks

NRM intensities for the sedimentary rocks range from 0.02 to 0.81 A m^{-1}, but little difference is seen between mudstone and arkose samples, with the mean for each group being 0.2 A m^{-1}. Initial directions for all sedimentary samples are

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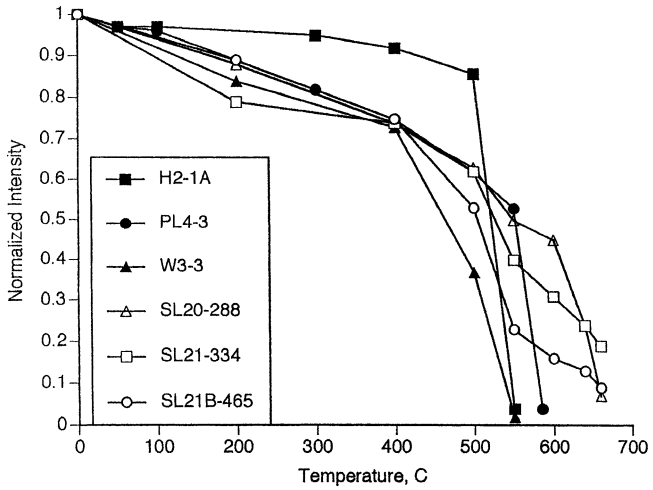


Figure 8. Normalized decay of the NRM during thermal demagnetization for the Holden (H), Pelham–Loudville (PL) and Ware (W) intrusions, mudstones (SL20-288 andSL21B-465) and arkoses (SL21-334).

well clustered, and give a mean direction close to, but slightly shallower than, the present magnetic field direction. Detailed thermal demagnetization studies reveal several separate components of magnetization, but with consistent demagnetization behaviour related to rock type. The arkoses (fine-grained sandstones) yield two components of magnetization (Fig. 7b). The initial component, A, seen from 0 to 200 °C or 400 °C, is steep and northeasterly. Above 400 °C, a second component B, with shallower, but still intermediate, inclination and northerly declination persists through higher temperatures to 660 °C. Mudstone samples have a more complex demagnetization behaviour, with both the A and B components visible. An additional component C, with shallow inclination, is present above 550 to 600 °C in nearly all of the mudstone samples (Fig. 7a). Intensity losses with demagnetization are plotted in Fig. 8, showing marginal loss of intensity up to 400 °C and then continued loss until 660 °C, when some (10–20 per cent) intensity is left. Both mudstone samples (SL20-288 and SL21B-465) and arkose samples (SL21-334) show a distribution of unblocking temperatures from 500 to 660 °C.

5 ROCK-MAGNETIC PROPERTIES

Petrographical observations were combined with measured rock-magnetic properties and demagnetization data to assess the behaviour and timing of acquisition of magnetization. Based on petrographical work, the main magnetic carrier in the diabases is titanomagnetite. In samples from the chilled margins of the intrusions, PSD-size titanomagnetite grains are present as skeletal grains, as rims or beads on the silicate grains, or as very fine elongate needles in the matrix. In samples from the interiors of the dykes there are more euhedral multi-domain-size titanomagnetite grains than in the contact samples, although there are also fine-grained titanomagnetites of varying morphologies in the matrix, as shown in the photomicrographs (Fig. 3). Although the petrographical work on the redbeds reveals several forms and generations of haematite as well as minor amounts of detrital magnetite, the predominant magnetic carrier is haematite.

5.1 Susceptibility values

The Holden, Ware and Pelham–Loudville diabase groups have average susceptibility values of 1.4×10^{-2}, 1.8×10^{-2} and 2.4×10^{-2} (SI units), respectively. The increasing susceptibility values follow the Fe-enrichment trend found in the diabase groups. The samples that are more iron-rich have higher CIPW normative concentrations of magnetite, correlating well with the measured susceptibility values and estimated modal amounts (McEnroe 1989). Minor but consistent susceptibility variations are found within each group. Chilled margin samples have lower susceptibility values than interior samples, and samples with maghemite have lower susceptibility values than samples with predominantly titanomagnetite.

Koenigsberger ratios (Q-values), when coupled with AF demagnetization spectra, yield information about magnetic stability and grain size. The Q-values range from 0.6 to 3.9 for Holden, 0.7 to 3.3 for Pelham–Loudville, and 0.3 to 2.7 for Ware diabases. The higher Q-values are usually from the chilled margins of the intrusive bodies. Q-values below 0.5 are considered to be dominated by multidomain remanence, and above 0.5 to be PSD to SD remanence (Stacey 1967). The AF behaviour, with MDFs above 20 mT, coupled with Q-values >0.5 indicate that the remanence is predominantly carried by PSD- to SD-size grains, in good agreement with petrographical observations.

Redbed samples fall into two distinct groups in susceptibility measurements, coincident with lithology. The mudstones have susceptibilities ranging from 3×10^{-3} to 6×10^{-5} SI, with a mean value of 9.3×10^{-4} SI. Susceptibilities of the arkose samples are at least an order of magnitude weaker, with a mean of 4.8×10^{-5} SI and a range of 3×10^{-4} to 2×10^{-6}. The susceptibility differences are consistent with the observed magnetic mineralogy.

5.2 Curie temperatures

Curie temperatures were measured on rock chips or powders, with the results listed in Table 1. The diabase samples have discrete Curie temperatures. A narrow temperature interval from 556 to 580 °C, indicating a near-end-member magnetite, was determined. Samples produce good reversible heating and cooling curves, indicating little oxidation or alteration of the titanomagnetite grains during the thermomagnetic runs. All measured samples contain observable titanomagnetite with oxidation-exsolution lamellae of ilmenite. The abundant ilmenite lamellae would account for the near-end-member compositions, because most of the titanium diffused into the ilmenite lamellae from the titanomagnetite during the oxidation-exsolution process. The range in Curie temperatures clearly indicates titanomagnetite as the predominant magnetic carrier. This conclusion is supported both by the optical observations and thermal unblocking temperatures.

Susceptibility versus temperature measurements on selected sedimentary samples reveal a gradual loss of susceptibility from 500 °C up to 670 °C, with marked changes in slope. In the mudstone samples two distinct Curie temperatures are inferred, namely a low-temperature fraction at 560 °C indicative of a low-titanium magnetite, and a higher-temperature fraction of 670 °C indicating haematite. Arkose samples tend to have very low bulk susceptibility values initially, and only indicate the presence of a pure (670 °C) haematite.

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Table 1. Curie temperatures and temperature susceptibility measurements.

Sample	Temperature
Curie temperatures on diabases	
H2	568 °C
PL9	556 °C
PL17	575 °C
PL19	562 °C
W3	580 °C
W7	567 °C
Temperature susceptibility on mudstones and arkoses	
SL20-288	560 °C, 670 °C
SL21-324	670 °C
SL21B-469	670 °C

5.3 Isothermal remanent magnetization (IRM)

IRM acquisition curves (Fig. 9a) show that 75 per cent or more magnetic saturation is achieved by 0.1 T in the Holden, Pelham–Loudville and Ware samples. The Holden and Pelham–Loudville samples achieve complete saturation by 0.2 T. Fine-grained samples from the Ware diabase do not saturate fully until 0.3 T, reflecting a higher-coercivity PSD component contributing to the overall signature. The complete saturation in all diabase samples by 0.3 T clearly indicates that titanomagnetite is the primary magnetic mineral, an interpretation supported by the demagnetization data and optical observations.

The medium destructive fields (MDFs) for the demagnetization of the IRM (Fig. 9b) are lower than the MDFs for AF demagnetization (Fig. 6), indicating a significant component of PSD behaviour (Lowrie & Fuller 1971; Bailey & Dunlop 1983). Generally, the MDFs for AF demagnetization vary from 20 to 40 mT, whereas only 10–15 mT was necessary to achieve an MDF for the IRM measurements. Fine-grained skeletal titanomagnetite crystals, and needles with a high shape elongation, are common crystal morphologies found in the matrix of the diabases and would contribute to the PSD behaviour. Overall most samples show a behaviour indicative of a mixture of PSD- and MD-size titanomagnetite grains.

Samples from the redbeds show distinctive saturation patterns related to their haematite content (Fig. 9a). Mudstone sample SL21-313 achieves 75 per cent saturation by 0.2 T, but does not fully saturate by 1.0 T, reflecting the mixed magnetite and haematite mineralogy observed in the sample. Other samples, both arkose (SL21-326) and mudstone (SL21B-467), show behaviour indicative of high-coercivity material and do not even begin to saturate by 1.0 T. This behaviour, combined with optical observations and demagnetization studies, indicates haematite only as the magnetic carrier. AF demagnetization of these same samples (Fig. 9b) corroborates this, with some loss of magnetization by the magnetite/haematite-bearing sample, but only minor reduction of magnetization in the haematite-rich rocks.

5.4 Hysteresis properties

Hysteresis properties were measured on selected diabase samples and representative samples of both mudstone and arkose. A typical hysteresis curve for a dyke sample shows complete

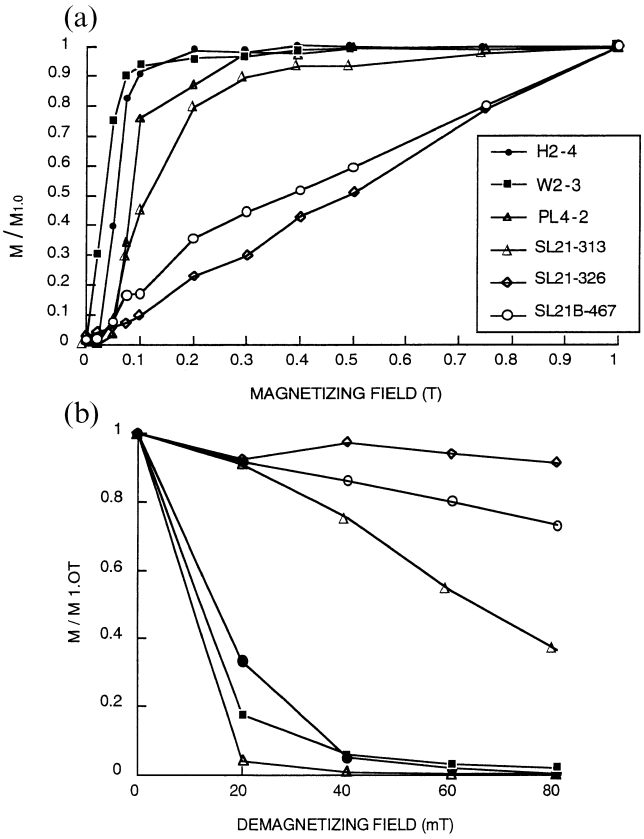


Figure 9. (a) Normalized IRM acquisition for selected samples in fields of up to 1 T. All diabase samples (H, W and PL) saturate by 0.3 T, while mudstone (SL21-313 and SL21B-467) and arkose (SL21-326) samples are only partially saturating by 1 T. (b) AF demagnetization of the saturation, or partial saturation, shown in (a). Diabases demagnetize at low fields with MDFs generally below 20 mT, whereas redbed samples only partially demagnetize, reflecting the significant haematite component in all samples.

saturation by 0.2–0.3 T, with a substantial remanence and coercivity for a diabase (Fig. 10c). The ratio of saturation remanence to saturation magnetization (M_{rs}/M_s) ranges from 0.2 to 0.45, and comparisons of coercivity of remanence to coercive force (H_{cr}/H_c) range from 1.3 to 2.7. In a Day plot of M_{rs}/M_s versus H_{cr}/H_c (Fig. 11), diabase samples plot in the PSD field with a trend heading towards the SD field. The Pelham–Loudville diabases have the highest M_{rs}/M_s values, usually above 0.3, and the Holden and Ware diabases are in the region of 0.2–0.35. The highest values are commonly from fine-grained chilled-margin samples. The trend towards the SD field is predominantly from the iron-enriched Pelham–Loudville group. All diabase samples have an H_{cr}/H_c ratio of less than 2. This predominant PSD component could be accounted for by the skeletal titanomagnetite morphologies, or the very fine needles and ultra-fine titanomagnetite dust commonly found in the matrix of the diabases.

Most of the mudstones and all of the arkoses show hysteresis behaviour requiring a 2.0 T field to saturate, and high coercivity, resulting in a lozenge-shaped curve (Fig. 10a). In some mudstone samples a large field (2.0 T) is still required to saturate, but a lower-coercivity phase is evident, giving the curve a classic 'wasp-waisted' geometry (Fig. 10b). These behaviours are entirely consistent with the other indicators of oxide mineralogy,

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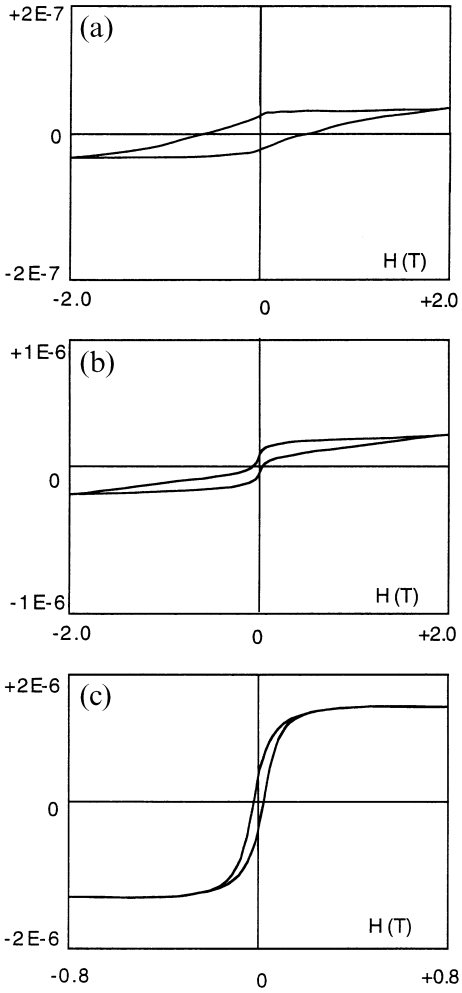


Figure 10. Typical hysteresis loops for (a) arkose and (b) mudstone of Fall River beds, and (c) diabases. The applied field was 2 T for sedimentary rocks and 0.8 T for diabases.

indicating that many samples are dominated by haematite, but that in some instances detrital grains of magnetite are present.

This predominance of fine haematite but some sparse detrital magnetite can be seen in the Day plot (Fig. 11). Several redbed samples have low H_{cr}/H_c values between 1.5 and 4.25 and corresponding M_{rs}/M_s ratios of less than 0.5, similar to hysteresis measurements on other haematite-dominated sedimentary rocks (Molina-Garza *et al.* 1996). One mudstone sample plotting out of this field, to the right of the SD field, with a high M_{rs}/M_s of 0.7, is known to have mixed mineralogy of fine crystalline haematite and detrital magnetite, as shown in Fig. 2(b). The SD nature of the haematite with a M_{rs}/M_s ratio of >0.6 and the detrital magnetite contribute to the H_{cr}/H_c value of 3.0. The arkose samples all have H_{cr}/H_c values >4.5, and two samples plot well beyond the limits of Fig. 11 (H_{cr}/H_c values of 22 and 24, respectively). Given that these hysteresis properties are bulk rock measurements, it is the coercivity of the whole sample that is measured and here reflects the mixed oxide population.

6 PALAEOMAGNETIC RESULTS

The site-mean directions and statistical parameters for both the three dyke groups and the sedimentary units are listed in

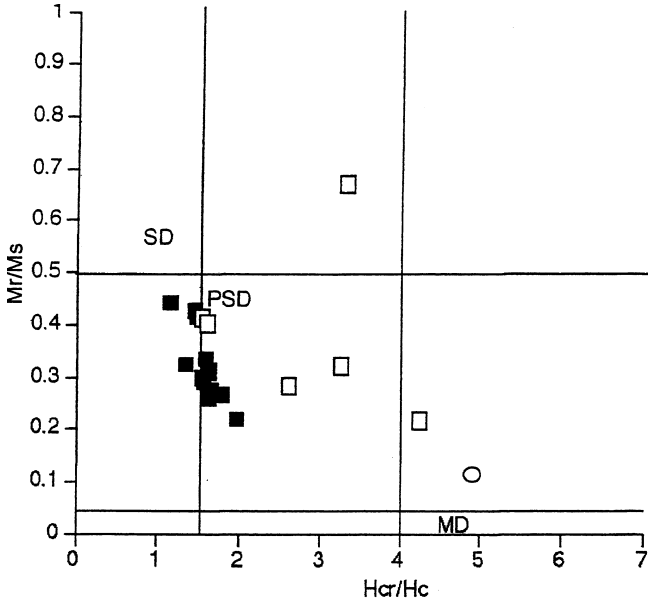


Figure 11. Day plot (Day *et al.* 1977) of hysteresis parameters of selected mudstones, arkoses and diabase samples. Symbols: diabases: filled squares; mudstones: open squares; and arkoses: open circles. Note: two arkose samples plot well off the diagram with H_{cr}/H_c values of 22 and 24, respectively.

Table 2. All samples measured have normal polarity, similar to results from other Newark Supergroup rocks of Early to mid-Jurassic age (Kodama & Mowery 1994; Kent *et al.* 1995). 34 of the original 36 dyke sites were used in calculating the Jurassic diabase palaeomagnetic poles, with two sites, H5 and P16, being omitted due to large scatter ($\alpha_{95}>20.0$). Magnetic directions for the diabases were calculated by averaging individual sample directions for each site and then averaging all sites for each of the three diabase groups as divided by chemistry and petrography, Holden, Pelham–Loudville and Ware. Each group yields a statistically distinct magnetic direction, with no overlap at the 95 per cent confidence level. The three groups show progressively steeper inclinations from the Holden ($D=13.8°$, $I=27.4°$, $\alpha_{95}=35.1°$) to the Pelham–Loudville ($D=5.4°$, $I=32.6°$, $\alpha_{95}=3.8°$) to the Ware ($D=6.9°$, $I=44.9°$, $\alpha_{95}=3.9°$). These mean directions are plotted in Fig. 12.

Mudstones from the earliest Jurassic Fall River beds yield a characteristic direction with shallow inclination that persists to high temperatures during thermal demagnetization, as well as two additional steeper components removed at lower temperatures. These two components are also seen in the coeval sandstones, without the presence of the high-temperature, shallow component. All palaeomagnetic directions seen in these Jurassic sediments are of normal polarity. Tilt corrections have been applied to both the C component of the mudstones and the B component of the arkoses, and, for reasons described below, the C component is assumed to have been acquired prior to tilting, while the B component was acquired after tilting. The resulting directions are: mudstones $D=14.0°$, $I=24.3°$, $\alpha_{95}=9.1°$; and arkoses $D=353.1°$, $I=47.7°$, $\alpha_{95}=5.9°$. The directions from the mudstones are concurrent with the Holden set of dykes, while the arkose data fall close to the mean direction of the Ware dykes (Fig. 12).

Sediments in the Deerfield Basin dip to the east towards the Connecticut Valley border fault. In the sections where the Fall

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Table 2. Palaeomagnetic directions and statistical parameters for diabases and sedimentary rocks.

Site	*N*/*No*	*D*	*I*	α_{95}	*K*	Plat °N	Plon °E	
Jurassic Diabases								
Holden Diabases								
H1	7/8	18.7	33.4	4.9	154	61.1	69.0	
H2	7/7	19.3	29.9	2.4	160	59.9	74.3	
H3	12/12	11.9	26.6	4.2	117	59.8	74.0	
H4	9/9	12.6	29.5	5.6	101	61.5	82.9	
H5*	7/8	2.9	23.0	20.0	10	59.6	102.4	
H6	8/8	14.3	23.6	10.0	32	59.9	84.5	
H7	10/15	6.7	21.1	10.4	23	58.1	95.5	
Site Averages	6/7	13.8	27.4	5.1	174	60.1	80.5	$A_{95}=4.1$
	7/7	12.2	26.8	5.8	137	60.3	83.3	$A_{95}=4.6$
Pelham–Loudville Diabases								
PL1	6/6	353.2	41.3	4.2	259	70.6	126.6	
PL2	6/6	6.7	31.8	10.0	46	64.3	93.1	
PL3	7/7	353.1	29.6	5.4	126	62.5	86.6	
PL4	3/3	10.0	30.1	6.9	316	62.5	86.8	
PL5	7/7	8.5	38.0	9.0	46	68.0	87.6	
PL6	6/6	0.4	28.3	5.6	143	62.4	106.7	
PL7	7/7	12.1	28.1	7.8	64	60.7	83.3	
PL8	17/17	4.3	26.0	4.1	78	59.0	121.7	
PL9	8/8	7.1	25.6	5.2	113	60.5	93.6	
PL10	6/7	14.1	24.6	12.3	40	59.3	80.5	
PL11	7/7	2.1	31.3	4.1	220	64.5	103.0	
PL12	5/5	6.3	34.4	16.9	59	66.1	93.2	
PL13	7/7	358.0	30.4	7.9	65	64.1	112.4	
PL14	6/6	9.5	32.8	5.6	142	64.3	86.5	
PL15	9/10	357.6	34.2	7.1	53	66.9	112.1	
PL16*	3/3	25.1	41.2	20.1	38	62.1	51.5	
PL17	3/3	5.6	34.4	7.6	264	64.8	116.1	
PL18	6/9	350.0	28.8	16.7	17	61.8	128.8	
PL19	8/8	356.3	32.0	3.7	225	64.8	116.1	
PL20	7/7	1.2	35.7	7.9	60	67.6	104.6	
PL21	5/5	39.4	33.8	12.3	40	49.1	40.7	
PL22	6/10	18.9	36.8	14.2	23	63.0	65.2	
PL23	4/5	6.9	41.0	8.0	133	70.3	88.2	
Site Averages	22/23	5.4	32.6	3.8	68	65.3	95.6	$A_{95}=4.1$
	23/23	6.2	33.1	3.9	62	65.5	93.7	$A_{95}=4.2$
Ware Diabases								
W1	6/6	13.3	45.8	5.5	151	71.3	68.2	
W2	5/5	15.6	45.0	2.5	917	69.7	64.2	
W3	6/7	7.6	46.3	7.1	90	74.0	82.9	
W4	5/5	358.0	45.5	5.8	115	74.5	114.7	
W5	8/10	4.2	43.8	3.2	308	73.1	94.9	
W6	4/4	8.4	47.7	7.4	154	75.0	78.0	
W7	5/6	1.7	39.4	7.1	116	69.7	103.5	
Site Averages	7/7	6.9	44.9	3.9	247	73.5	85.8	$A_{95}=3.9$
Sugarloaf Formation								
Mudstones—C component								
Uncorrected	30/34	4.8	13.5	9.1	9	54.0	99.4	dp = 4.7 dm = 9.3
Tilt corrected	30/34	14.0	24.3	9.1	9	57.7	81.3	dp = 5.2 dm = 9.7
Arkoses—B component								
Uncorrected	36/39	353.1	47.7	5.9	18	75.1	131.6	dp = 5.0 dm = 7.7
Tilt corrected	36/39	35.9	45.3	5.5	20	56.9	34.3	dp = 4.7 dm = 7.0

N/*No*: ratio of samples used in determining site mean directions to number of samples demagnetized; *D*, *I*: mean declination and mean inclination, in degrees; α_{95} and *K*: statistical parameters; Plat °N and Plon °E: latitude north and longitude east of magnetic pole; *selected sites not included in first site averages.

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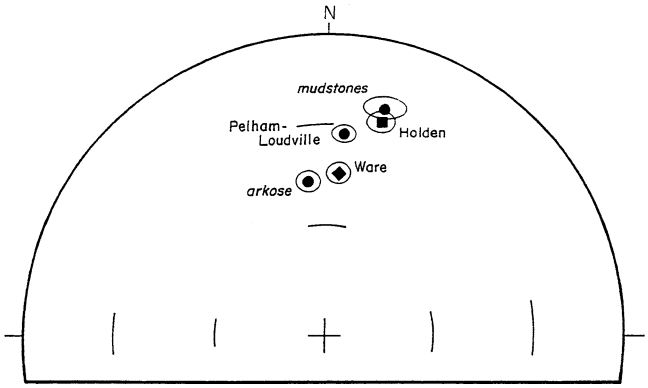


Figure 12. Equal-area projection showing characteristic mean magnetic directions and associated 95 per cent confidence circles for Holden dykes (filled square), Pelham–Loudville (filled circle), and Ware intrusions (filled diamond), mudstones (filled circle) and arkoses (filled circle).

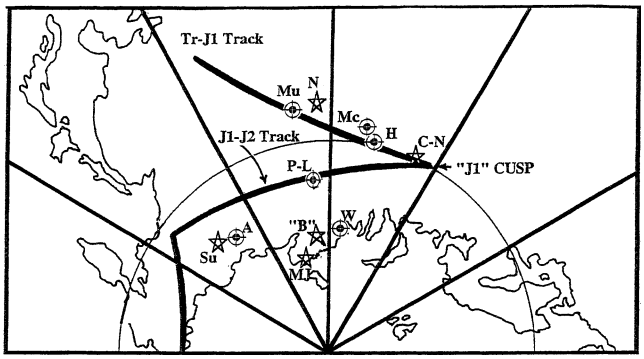


Figure 13. A Mesozoic APWP for North America from May & Butler (1986). Plotted from this study are: H, the Holden diabase; P-L, Pelham–Loudville diabases; W, Ware diabases; Mc, Fall River mudstones corrected for tilt, and Mu, uncorrected; and A, the arkoses. These are compared with selected published poles of Jurassic age, shown as stars: N, the Newark Hettangian pole (Witte & Kent 1990); C-N, Culpeper and Newark basin igneous units and baked sediments (Kodama *et al.* 1994a); "B", remagnetization component from the Newark Basin (Witte & Kent 1991); Su, Sassamansville diabase (Kodama & Mowery 1994); and MJ, calculated Middle Jurassic pole for Gondwana (Van Fossen & Kent 1990).

River beds were sampled, dips of the bedding range from 25° to 40°, with two sections having strikes of N10°E and the third short section having a strike of N60°E. The results of mean palaeomagnetic directions for the two rock types before and after tilt corrections are given in Table 2. A fold test on both the C and B components proved to be inconclusive, with little or no change in precision with untilting. This is due to both the scattered nature of the data initially and the similarities in tilt parameters for most of the samples.

Mean palaeomagnetic poles have been calculated for the three diabase groups and the two redbed lithologies and are included in Table 2. Two of the diabase groups, the Holden group (60.1°N, 80.2°E, $A_{95}=4.1°$) and the Pelham–Loudville group (65.3°N, 95.6°E, $A_{95}=4.1°$) fall very near May & Butler's (1986) Mesozoic APWP for North America (Fig. 13). The Holden group pole lies on the part of the track before the J1 cusp, while the Pelham–Loudville pole falls after the cusp. The characteristic direction of the mudstones in their pre-tilting position yields a pole of 57.7°N, 81.3°E, lying close to the Holden group pole on the pre-cusp side of the APWP. The third diabase group, the Ware dykes, has a mean pole of 73.5°N, 85.8°E ($A_{95}=3.9°$), and lies well north of the Jurassic part of the APWP (Fig. 13). A palaeomagnetic pole for the B component of the sediments, represented by the arkose samples, also lies at high latitude (75.1°N) but significantly east (131.6°E) of the Ware group pole.

7 DISCUSSION

7.1 Magnetization—diabases

Evaluation of the rock-magnetic properties, the thermal AF demagnetization and optical observations leads to the conclusion that many of the remanent directions in the diabases were obtained upon cooling through the Curie temperature for PSD-size titanomagnetite grains, and should be considered a TRM or CRM. Titanomagnetite was modified by a high-temperature oxidation, forming ilmenite oxidation-exsolution lamellae. Based on the Curie temperatures and unblocking spectra, the magnetite has an ulvöspinel component of approximately 0–15 per cent. If this oxidation occurred above the Curie temperatures, 556–580 °C, then these directions would be considered a TRM, but if this oxidation continued below the Curie temperature then the remanent directions should be considered a CRM. The oxidation-exsolution textures argue for a TRM for many samples. The TRM would have been acquired during magmatic cooling, after oxidation-exsolution, which mostly would have occurred between 850 °C and 600 °C. Most of the coarsening of the oxidation-exsolution lamellae would have occurred above the consolute point for magnetite-ulvöspinel at approximately 600 °C (Buddington & Lindsley 1964). Significant oxygen mobility in magnetite requires temperatures well above 300 °C (Valley & Graham 1991), and therefore the reworking of titanomagnetite grains by coarsening of oxidation-exsolution ilmenite lamellae, or forming of new lamellae, would cease well above this limit. A more limiting factor may be the mobility of Ti and Fe in the magnetite structure at higher temperatures.

In contrast to the above scenario there is a subset of dyke samples within the Pelham–Loudville group which show a lower-temperature oxidation feature in which a 'lit-par-lit texture' is common. This texture is interpreted to be the product of intermediate-temperature exsolution of ulvöspinel from magnetite, forming below the consolute point for magnetite-ulvöspinel at about 600 °C. Subsequently, this was followed by a lower-temperature oxidation forming ilmenite from ulvöspinel. These samples may be considered to carry a CRM, because the Curie temperature is above the exsolution temperature. Ware and Holden diabase samples that show a granulation to the magnetite or maghematite would also carry a CRM. Based on the silicate and oxide petrology, the granulation and maghematization probably occurred as a deuteric alteration during cooling of the diabases. Within each diabase group, the remanent directions from the samples/sites that underwent this true exsolution, and those that have higher-temperature oxidation-exsolution features or minor maghematization are similar at the 95 per cent confidence level. In either case the magnetization appears to have been obtained during or shortly after emplacement of the dykes.

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Palaeolatitude determinations for the three dyke groups indicate subtropical positions of 14° for the Holden group, 17° for the Pelham–Loudville group, and 26° for the Ware group, similar to positions determined from sediment magnetism (Kent *et al.* 1995). Dispersion of site-mean VGPs for the Pelham–Loudville group ($S = 10.3°$) is consistent for low-latitude geomagnetic secular variation. Both the Holden and the Ware groups yield lower angular dispersion (5° to 6°), but are not inconsistent with other dispersion measurements for Mesozoic dykes (Van Fossen & Kent 1993). Some averaging of the field must have occurred, given that the width of several of the dykes is greater than 50 m and that they cooled relatively slowly at depth. It would be fortuitous for the poles of the Holden and the Ware groups to fall near published poles of similar ages if they only represent spot readings of the magnetic field, without having averaged out secular variation.

Although there is eastward tilting of Mesozoic units within the Hartford and Deerfield basins due to motion on the border fault (Robinson *et al.* 1986), there is little evidence of regional tilting outside the rift basins. No systematic or significant variation in palaeomagnetic directions is observed between the Loudville dykes, cropping out west of the basin margin, and the Pelham dykes, found to the east in the metamorphic highlands. If tilting did occur along the eastern side of the border fault it is not recognizable within dyke groups, as directions are similar and independent of geographical location.

7.2 Magnetization—sedimentary rocks

The sedimentary samples all yield optical observations, thermal demagnetization, IRM, intensity versus susceptibility, and hysteresis measurements that indicate that haematite is the major iron oxide present. This is strongly supported by their obvious red colour. The extensive microscopy, plus hysteresis curves and IRM acquisition data indicate that haematite is the predominant remanence carrier, although it can be found in a number of forms and grain sizes. The distributed unblocking temperatures on thermal demagnetization (Fig. 8) imply that a combination of fine pigment and larger grains of haematite (specularite) are present (Collinson 1974). Other observations from optical and rock-magnetic work agree that rare detrital grains of magnetite are present, although most detrital magnetite grains have altered to martite.

Up to three components of magnetization are observed in the redbed samples, and these components can be related to rock type and specific geological histories. All samples show a low-temperature steep overprint, closely related to the present-field direction. Arkose samples have only two components of magnetization, namely the present-field overprint and the B component. These rocks are characterized by a predominance of quartz and feldspar grains with only a small amount of Fe-silicates. It appears that detrital and early authigenic haematite were bleached out by later fluids (Wise *et al.* 1992; Hubert & Dutcher 1999), leaving the rocks predominantly grey-buff in colour. The small amount of iron oxide now found in the arkoses was deposited on top of dolomite cement, late in the diagenesis of the rocks (Hubert & Reed 1978), indicating the magnetization was acquired well after deposition. The fine-grained mudstones yield both B and C components, with the high-temperature, shallow C component interpreted as the characteristic direction. These rocks contain martite, specularite, haematite staining from the dehydration of detrital limonite stains, and rare detrital magnetite, all of which contribute to the observed remanent magnetization. The detrital grains as well as the early authigenic haematite are responsible for the high-temperature C component. Later alterations, such as intra-stratal solution of Fe-silicates forming ultra-fine haematite, provide carriers for the steeper B component. A proposed middle Jurassic hydrothermal event (Sutter 1988; Roden & Miller 1991) may be responsible for the bleaching of the permeable arkoses, thus removing the material carrying the primary direction, while the more impermeable mudstones remained intact.

In accordance with similar studies in Newark Supergroup sediments (Witte & Kent 1990; Kent & Olsen 1997), the shallow, high-temperature component is considered to be characteristic, and to have been acquired during or soon after deposition. The steeper lower-temperature component seen in some of the mudstones and all of the arkoses is assumed to have been acquired after tilting in mid-Jurassic times, related to a thermal event and fluid migration at around 175 Ma. (Sutter 1988; Witte & Kent 1991; Hubert *et al.* 1992). These assumptions are not reinforced by a positive or negative tilt test, so it is possible that the C component could have been gained during the tilting, known to be just post-deposition of the sediments (Stevens & Hubert 1980; Hubert & Dutcher 1999). The B component must have been acquired after the basin tilting, as the remanence carriers were deposited by migrating fluids late in the diagenesis of the bleached arkoses.

7.3 Redbed/dyke pole comparisons

The three distinct pole positions from the dyke groups and the poles from the two sedimentary components define a range of directions around the Mesozoic APW path (Fig. 13). The C component of the mudstones prior to tilting and the Holden dyke poles are similar, and both fall on the latest Triassic–earliest Jurassic part of the track, well before the J1 cusp. This is to be expected for the Fall River beds deposited at the Triassic–Jurassic boundary, with a C component acquired during or soon after deposition. This pole falls between a pole for Newark sediments of Hettangian age in the Newark basin (55°N, 95°E; Witte & Kent 1990) and a pole for baked sediments and igneous units in the Culpeper and Newark basins, dated at 201 Ma (60°N, 69°E; Kodama *et al.* 1994a). The data presented here from sediments deposited at the Triassic/Jurassic boundary and a dyke set of presumed Early Jurassic age strengthen the case for the presence of the J1 cusp and the notion of rapid plate motion at this time, in agreement with other Early Jurassic studies (Kodama *et al.* 1994a). Other palaeomagnetic studies (Kent & Witte 1993; Kent *et al.* 1995) indicate small variations in pole positions during the late Triassic–earliest Jurassic, implying slow rates of plate motion.

The Holden dyke group plots close to the C component on the APW path (Fig. 13), prior to the J1 cusp, while the Pelham–Loudville group pole lies on the APW path well after the J1 cusp. Correlation with other igneous units, particularly dykes and sills in the Hartford Basin, is difficult due to age uncertainties and palaeomagnetic unreliability of earlier published data (Smith & Noltimier 1979). Although there are no radiometric data from either of these dyke groups, the differences in their chemistries suggest that they represent

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different times of igneous activity. The Pelham–Loudville dykes agree with palaeomagnetic data from lava flows in the Hartford Basin (Prevot & McWilliams 1989), although the correlation of dyke groups to specific lava flows is not as obvious as suggested by Philpotts & Martello (1986).

The Ware diabases yield a mean pole at 73°N, 86°E, north of the APW path of May & Butler (1986) for Early to mid-Jurassic. This same high-latitude pole position has been reported in the chemically correlative Buttress Dyke in Connecticut (75°N, 92°E; Smith & Noltimier 1979) and in the 'B' overprint component (74°N, 96°E) found in the Newark Basin sedimentary rocks (Witte & Kent 1991). Witte & Kent (1991) use a 'working age' of 175 Ma for this overprint by relating its generation to a possible thermal event as suggested by disruption in the argon system in the potassium feldspars (Sutter 1988). Reported $^{39}Ar/^{40}Ar$ ages of 176 ± 3.8 and 181 ± 6.7 Ma for the Buttress dyke (Sutter & Smith 1979) imply that the Ware–Buttress dykes are among the youngest of the intrusive units in the Newark Supergroup. Although this dyke group has been suggested as a feeder dyke to the Holyoke Basalt (Philpotts & Martello 1986), palaeomagnetic, geochemical and field relationships make this correlation difficult. The high-latitude poles seen in the Ware and the Buttress dykes are never seen in the Holyoke Basalt, and perhaps more significantly, the Holyoke Basalt is characterized by shallow positive and negative inclinations (Prevot & McWilliams 1989; Hozik 1992) in sharp contrast to the Ware–Buttress directions.

We can compare these high-latitude poles to a mid-Jurassic Gondwana pole (173 Ma) determined by Van Fossen & Kent (1990) as part of their study on the Moat volcanics in New Hampshire. This pole (76°N, 109°E) is at high latitude, and a similar longitude to both the B component and the Ware dykes. The agreement of a high-latitude pole from the Newark B component, the Ware dykes, and a mean Gondwana pole underscores the need to reassess the APWP for the Middle Jurassic.

In this study, the B component of the arkoses also yields a high-latitude pole (72°N, 132°E), but one that is considerably farther east than that of the Newark Basin B component (Fig. 13). The age of this overprint appears to be younger than the 175 Ma suggested for the B component in the Newark Basin. Evidence supporting this is found in reset fission-track ages of 118–130 Ma (Roden & Miller 1991) in the Deerfield Basin, which indicate that closure temperatures for apatite did not occur until the Middle to Late Jurassic (Roden & Miller 1991). Coincident with these elevated temperatures was extensive hydrothermal activity and fluid migration (Hubert & Dutcher 1999), providing for the bleaching of the arkoses and production of new ferric iron phases. The interbedded mudstones were generally impermeable to extensive bleaching (Stevens & Hubert 1980; Hubert & Dutcher 1999), and, although they experienced some alteration, were able to retain their original component of magnetization. The arkose B component is not unlike a remagnetized direction in the Sassamansville Diabase (72°N, 137°E) determined by Kodama & Mowery (1994) and a B component seen in the Passaic sediments in the Sassamansville syncline of 67°N, 147°E (Kodama *et al.* 1994b). It must be noted that, if the site locations for these sample are rotated some 15°, the directions move into coincidence with other Newark Basin overprint directions, as suggested by Kodama *et al.* (1994b).

Palaeomagnetic directions for some of the dykes and sediments in central Massachusetts agree well with proposed APW paths for North America in the Jurassic. As can be seen in Fig. 13, the C component of the mudstones and the primary directions in the Holden and Pelham–Loudville dyke groups fall on or close to the APW path, either before or after the J1 cusp. The Ware dykes and the B component of the sediments both indicate higher-latitude pole positions than predicted by the May & Butler (1986) APWP. Earlier APW paths, constructed from averaging individual palaeopoles, also show higher-latitude positions during the Middle Jurassic (Irving & Irving 1982). This controversy has been explored by looking at data from other continents, either by using palaeopoles from tectonic regions external to North America (Van der Voo 1992) or by producing synthetic paths for North America using data from other Atlantic-bordering continents (Courtillot *et al.* 1994). These studies produce Jurassic APW paths at either 70° or 75°N latitude, but both are hampered by limited data. Our study provides additional poles from both igneous and sedimentary rocks confirming the high-latitude aspect of the APWP of North America during the Jurassic.

8 CONCLUSIONS

A palaeomagnetic study of igneous and sedimentary rocks from central Massachusetts produces several Early to Middle Jurassic pole positions that confirm the J1 cusp and substantiate a high-latitude Middle Jurassic pole. The Early Jurassic pre-J1-cusp pole positions are carried by a within-plate type of tholeiite, the Holden diabases, and lacustrine sediments at the Triassic–Jurassic boundary. The lacustrine mudstones have a high-coercivity component with peak unblocking temperatures of 660 °C, and a shallow normal polarity with northeasterly direction, interpreted as a primary direction carried by early haematite. A steeper component found in the mudstones and arkoses was revealed above 400 °C. This component is of normal polarity with north-northwesterly direction, predominately carried as a secondary haematite present as a cement in the arkose and as a breakdown product of the Fe-silicates in the mudstones. The high-latitude pole of this magnetization is similar in latitude to the B component seen in the Newark sediments.

The three chemically and petrographically distinct igneous units produce three well-isolated palaeopoles. The oldest igneous phase, the HTQ Holden diabases, have a peak unblocking temperature of 570 °C and carry a shallow normal polarity with northeasterly direction coincident with the primary direction in the mudstones. The post-J1-cusp HFQ iron-rich Pelham–Loudville diabases carry a characteristic component of normal polarity with a northerly direction and are steeper than the Holden diabases. The distinct low-titanium LTQ Ware diabases carry a steep normal polarity with a northerly direction. This direction is very similar to the 'B' component in the Newark Basin (Witte & Kent 1991), and in inclination to the Sassamansville Diabase (Kodama & Mowery 1994) and a 'B' component seen in the Passaic sediments in the Sassamansville syncline (Kodama *et al.* 1994b). It is possible that the Ware intrusions represent an episode of igneous activity at approximately 175 Ma which provided the necessary heat for the hydrothermal alteration that may have produced the pervasive remagnetization giving the high-latitude poles found in the Newark, Culpeper, and Hartford–Deerfield basins.

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ACKNOWLEDGMENTS

We gratefully acknowledge the facilities and helpful personnel at the Institute of Rock Magnetism, University of Minnesota, John Hubert for discussion on the sedimentary history and petrography of the Deerfield–Hartford Basin, and Peter Robinson for discussion and comments on the structural history of central Massachusetts. Thoughtful reviews by Ken Kodama and John Stamatakos improved the manuscript.

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